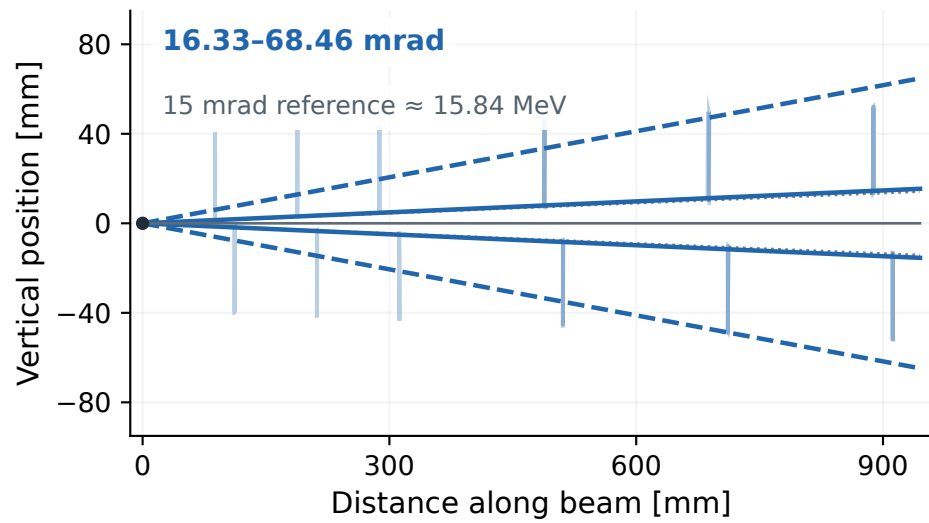


HPS geometry, straight-track acceptance and selected spectra | v5.7.1

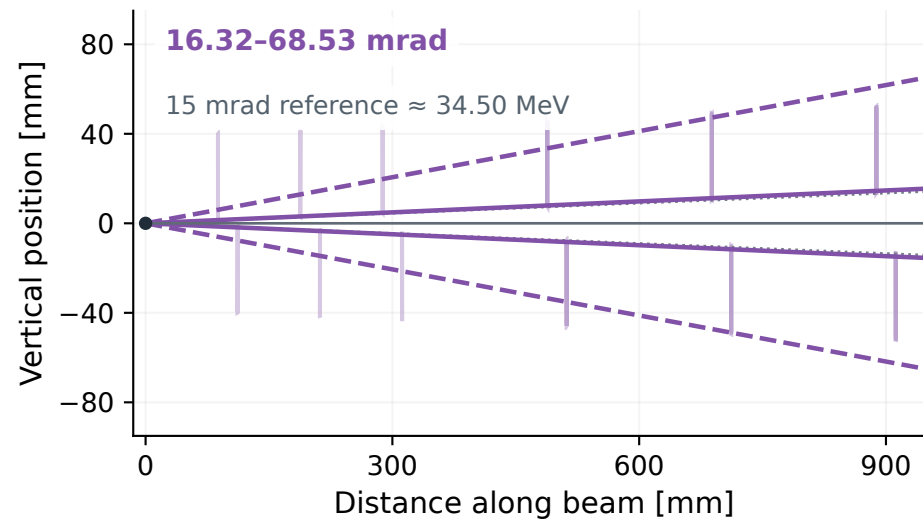
A | Sensor layout in the vertical projection

HPS 2015 | 1.056 GeV



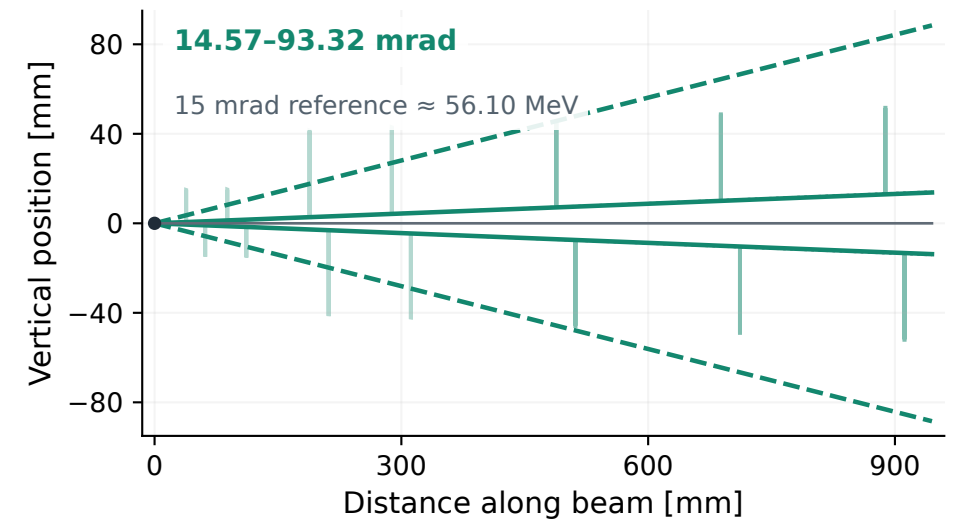
Projected mass scale: 17.27-72.24 MeV

HPS 2016 | 2.300 GeV



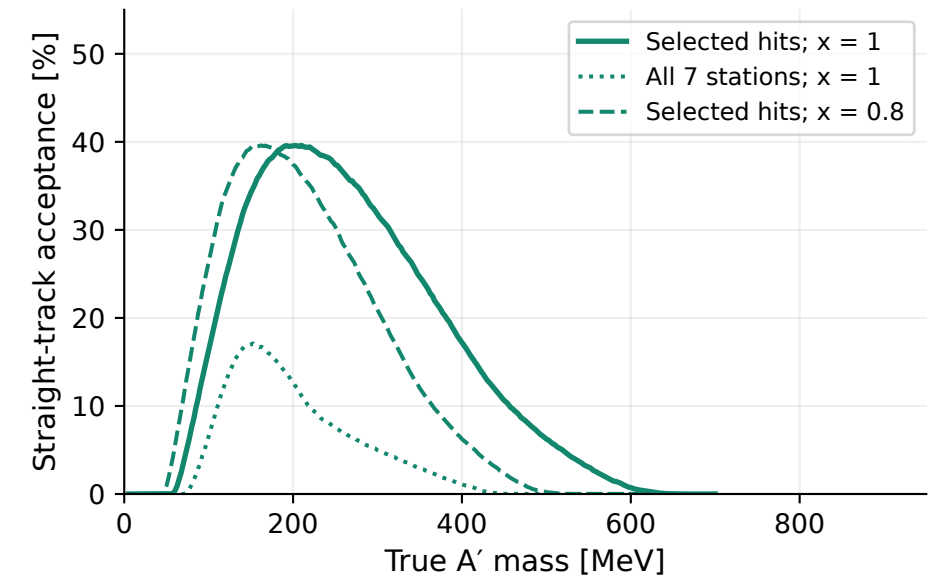
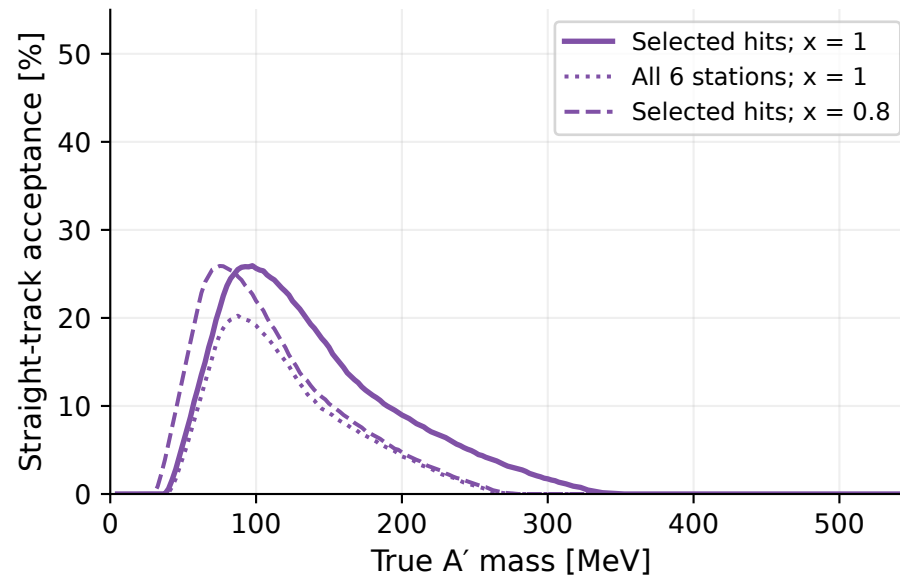
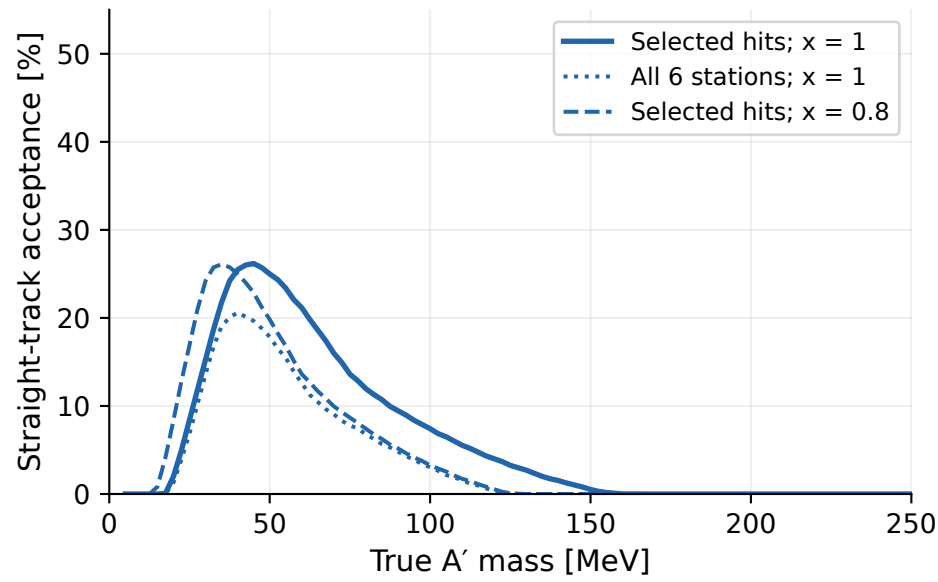
Projected mass scale: 37.54-157.50 MeV

HPS 2021 | 3.740 GeV

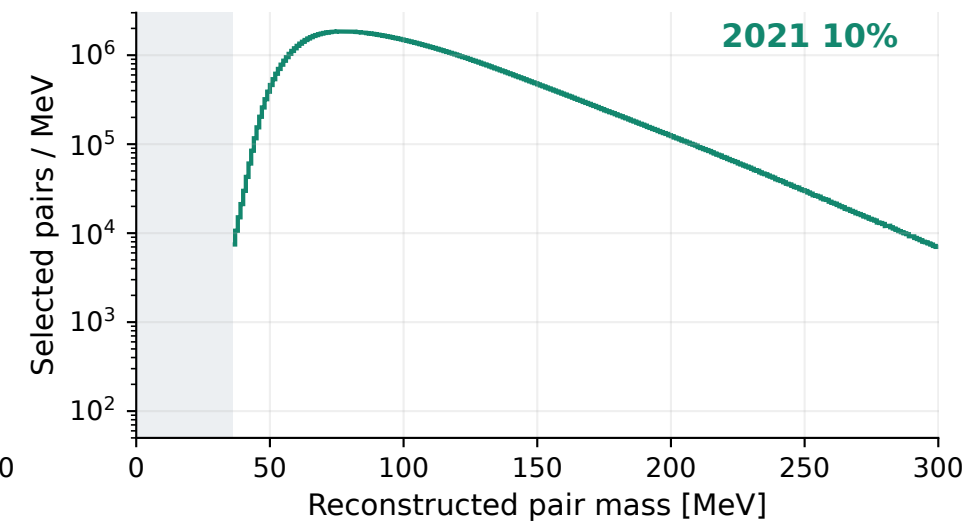
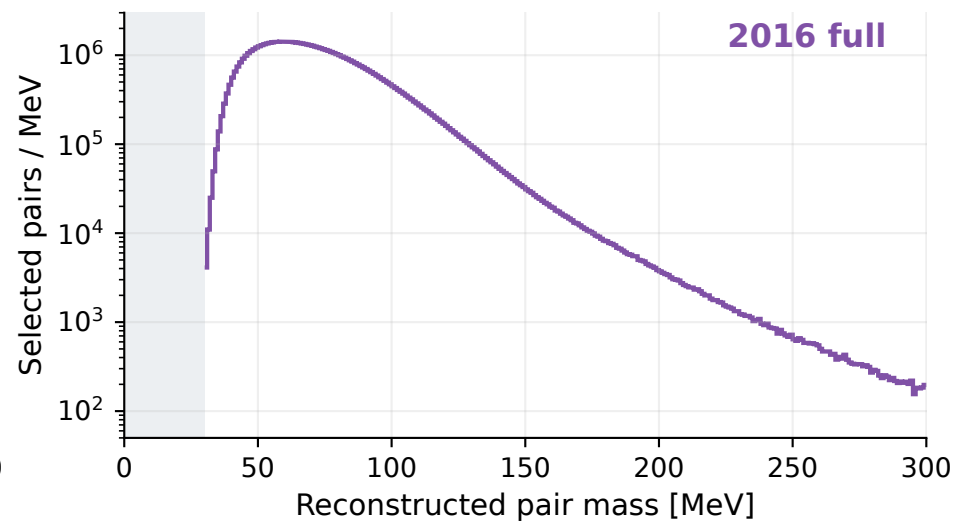
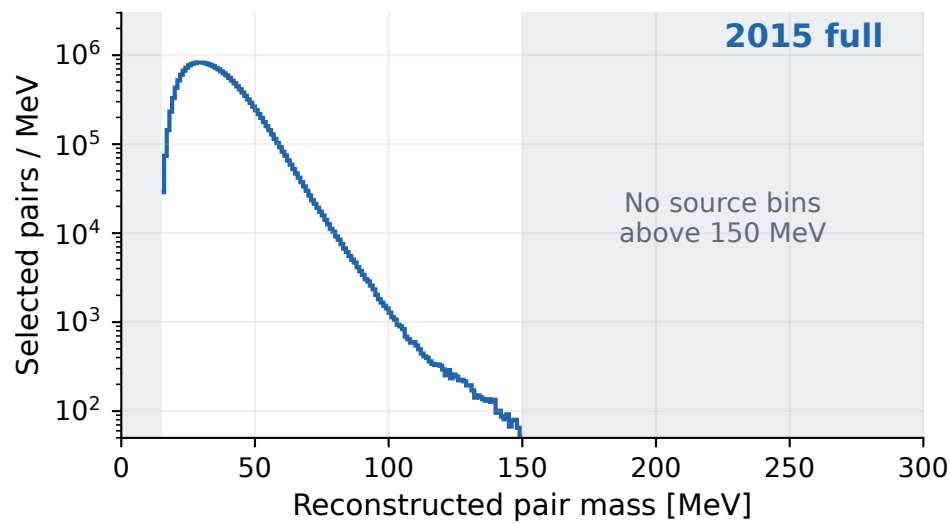


Projected mass scale: 54.50-348.53 MeV

B | Run-dependent hit requirements; $x = E(A')/E(\text{beam})$



C | Existing selected-pair spectra, without exposure scaling



Hit cuts: 2015 ≥ 5 paired stations each + e^+ L1/L2; 2016 ≥ 5 each; 2021 ≥ 10 e^+ / ≥ 8 e^- strip hits.

A shows projected mass scales, not 3D endpoints. B retains finite faces and beam rotation; field, material, trigger and reconstruction are omitted.